Interlinkages and Price Dynamics among Gold ETFs, Sovereign Bonds and Physical Gold in India

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Granger causality,
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ABSTRACT

This study examines the comparative risk–return characteristics and resilience of three key gold investment instruments in India over the volatile period from 2015 to 2025. Using monthly data, econometric tests such as correlation, regression, causality, and cointegration analyses are used to evaluate the short and long-run interrelationships. The findings show that while all three move broadly together, their return dynamics differ significantly in stability, responsiveness, and efficiency. SGBs are the most stable and rewarding in the long term, ETFs as the most market-responsive, and physical gold as the least efficient form of gold investment. The findings reveal a crucial dichotomy: while daily returns are largely uncorrelated, in the long run they can be used interchangeably.

1. **Introduction**

Gold has held a unique place in India both culturally and economically. It has been utilized as a hedge against uncertainty and inflation for years. In addition to physical gold, new alternatives for gold investment have been introduced such as Gold ETFs (Exchange Traded Funds) and Sovereign Gold Bonds (SGBs). SGBs are issued by the RBI and backed by the Government. They were introduced by the government in 2015. They provide 2.5% annual interest rate (paid semi-annually) and are completely tax-free if held till maturity (8 years). ETFs, on the other hand, are traded in the stock market just like stocks. They are highly liquid and flexible, due to which active traders prefer ETFs as compared to long term investors who prefer SGBs. The price mimics the actual gold prices and has no storage costs and neither do they pay any interest. Both these instruments are ideal for reducing our country's growing dependence on imports to meet our gold demand.

This study uses empirical data from 2015 to 2025 to analyse the relative returns and linkages between these investment instruments. Demonetization , the introduction of the Goods and Services Tax (GST), the COVID-19 pandemic, supply-chain disruptions and inflationary pressures led to the extreme macroeconomic volatility during those years. This paper examines the empirical relationships between these three gold instruments. In particular, the research aims to answer two fundamental questions:

i. Do the daily returns of these instruments show a short-term linear relationship?
ii. Do the prices of these instruments maintain a stable, long-term equilibrium?

2. A first look at the data

This analysis mainly is concerned with the following:

- Sovereign Gold Bonds
- Gold ETFs
- Physical Gold

Although the study aims to compare physical gold, SGBs, and Gold ETFs, the long-term and standardized form of gold's spot price in India is not available for the time period we have chosen. We use daily gold futures prices (in USD) and the exchange rate downloaded directly from Yahoo Finance, for the period January 2015 to October 2025 and then converted the gold prices from USD to INR. Then, monthly percentage returns were calculated from the resulting series to analyse gold's risk–return behaviour over time. The analysis includes summary statistics and time-series visualizations depicting both price trends and return volatility. Summary statistics can be observed in Table 1.

Table 1.
Summary statistics

	SGB Return	ETF Return	Gold Return
count	95.00	95.00	95.00
mean	1.674	1.197	1.617
std	4.057	4.728	3.782
min	-6.250	-22.927	-6.180
25%	-0.790	-1.339	-0.882
50%	1.010	1.255	1.414
75%	3.665	3.891	4.383
max	17.466	13.313	11.972

Table 1 indicates that all three gold forms show positive average monthly returns. SGB and Physical Gold slightly outperform Gold ETFs. Gold ETFs exhibit the highest volatility implying more month-to-month fluctuation, while Physical Gold is the most stable. SGBs offer the best average return with moderate volatility, indicating a relatively better risk-adjusted performance. ETFs are much riskier whereas SGBs and physical gold are comparatively more stable. On the other hand, SGBs reach the highest monthly peak.

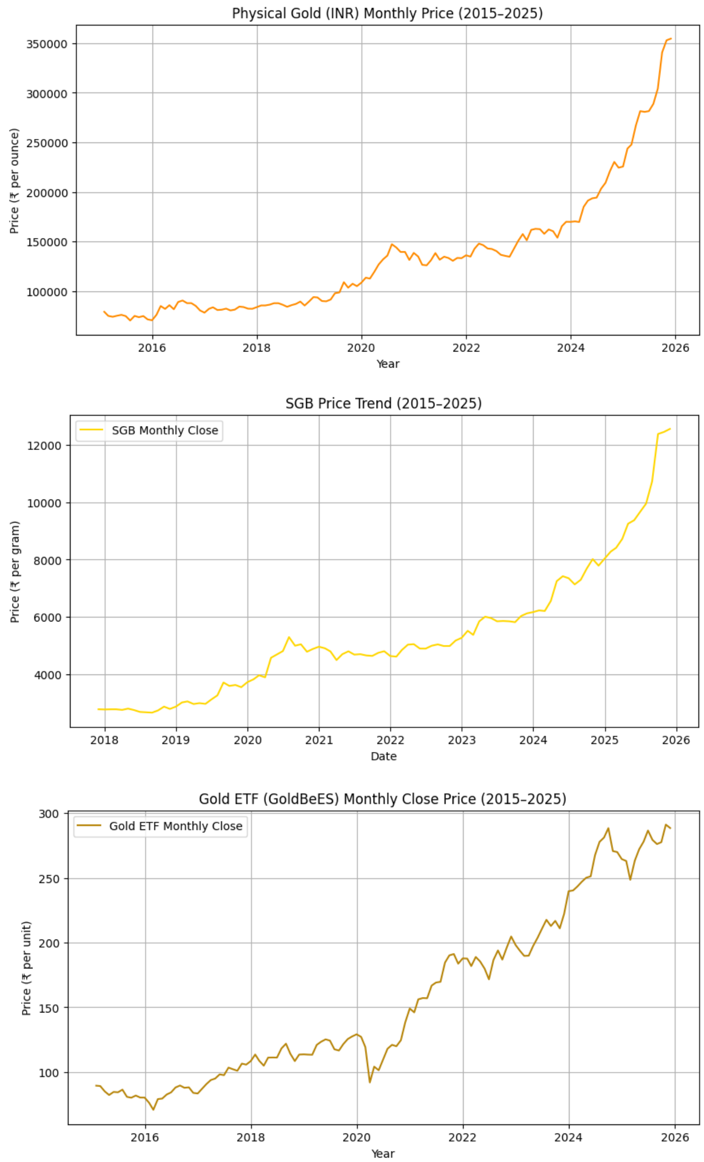


Fig 1. Monthly returns of SGB, ETF & Physical

Fig 1 indicates the long-term price movement of all three instruments : SGBs, Gold ETFs, and physical gold. The figure shows that all three instruments showcase an upward trend which indicates the global rise in gold prices driven by multiple macroeconomic events such as the 2020 pandemic-induced uncertainty, introduction of GST, demonetization and post-2022 inflationary pressures following global monetary tightening. While the overall trend is similar across instruments, the SGB price series is comparatively smoother and less volatile whereas Gold ETFs and physical gold show greater sensitivity to daily global shocks. This stability is due to the interest-bearing feature of SGBs, while the sensitivity reflects real-time pricing on the NSE and MCX platforms. The upward convergence after 2020 emphasizes gold's importance as a safe-haven asset in the face of rising global danger.

Fig 2. Rolling 12-month volatility for all three instruments

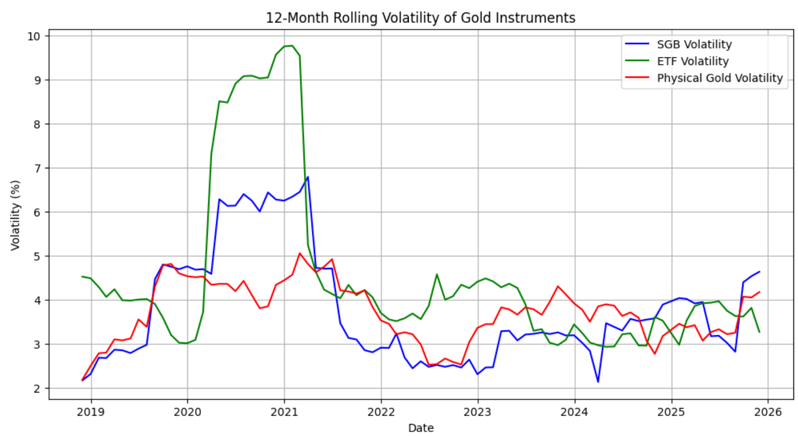


Figure 2 illustrates the rolling volatility dynamics of all three instruments. It shows how the risk of SGBs, Gold ETFs, and Physical Gold has evolved from 2015 to 2025. Volatility surged sharply during 2020–2021 due to the COVID-19 crisis, which is most evident in ETFs and to a lesser extent in SGBs. It can be seen that volatility gradually declined in all three instruments. But in the long run, it can be seen that ETFs tend to be the most volatile likely due to its exposure to daily market trading activity, physical gold moderately volatile and SGB's, the least volatile attributed to its interest-bearing nature.

3. **Methodology**

For the econometric analysis, two datasets were derived:

a. **Daily Returns Data:** Calculated as the daily percentage change in price, multiplied by 100 (%Return = $\frac{P_t - P_{t-1}}{P_{t-1}}$ x 100). This data was used for descriptive statistics and OLS regression.

b. **Price Index Data:** A normalized Price Index was created for each series. Price Index$_t$ = Price Index$_{t-1}$ x $\left(1 + \frac{Return_t}{100}\right)$. This non-stationary series is the necessary input for cointegration testing. The three resulting series are named : SGB_Price_Index, ETF_Price_Index, and Gold_Price_Index

The following econometric tools were applied:

- **Descriptive statistics** : Descriptive statistics were computed for the daily returns to characterize the risk and return profiles of the Gold ETF (etf_ret) and Physical gold (gold_ret).

- **Correlation and regression** : To examine the daily, short-term relationship between the two most liquid instruments, an OLS regression was performed with ETF returns as the dependent variable and Physical gold returns as the independent variable.

$$etf\_ret_t = \beta_0 + \beta_1 \cdot gold\_ret_t + \epsilon_t$$

The goal was to determine the degree to which daily movements in the highly liquid physical gold market predict or correlate with the daily movements of the ETF.

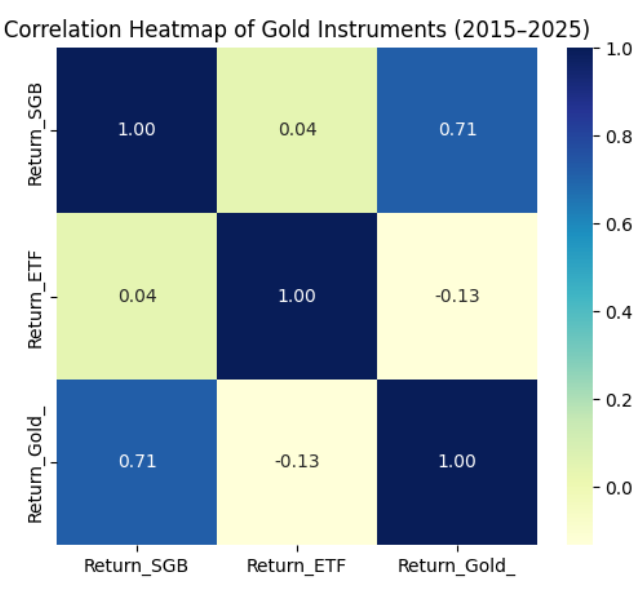
- **ARCH and ADF tests** : The Augmented Dickey–Fuller (ADF) test was used to check the stability of return data, ensuring consistency over time whereas the ARCH test helped identify periods of volatility clustering. These tests together confirmed the reliability of the dataset for examining gold instruments' performance and risk behaviour.

- **Granger causality** : The Granger causality is used to check if changes in gold prices can predict movements in SGB or ETF returns. It examines if gold influences these instruments in the long run.

- **Johansen cointegration :** The Johansen cointegration test checks whether SGBs, Gold ETFs, and physical gold move together in the long run despite short-term fluctuations. It helps determine if these assets share a stable long-term equilibrium relationship.

4. Results

4.1 Correlation



The heatmap shows the correlation between the three gold investments. SGBs and physical gold have a correlation of 0.71 which indicates that they move in the same direction, showing a strong link between the two. In contrast, Gold ETFs have a negative correlation of -0.13 which implies they behave quite differently. This means while SGBs mirror real gold's performance fairly well, ETFs tend to react more to short-term market factors and trading activity. The weaker link between SGBs and the other two instruments comes from their limited trading activity and interest-bearing nature, which smooths out daily price swings. This stability means that including SGBs in a gold-focused portfolio can slightly lower overall risk without sacrificing much return.

4.2 Regression Analysis

A primary part of the analysis involves regressing ETF returns and SGB Returns against gold returns to determine their statistical relationship:

- Dependent Variable: ETF Returns & SGB Returns
- Independent Variable: Gold Returns

Model 1: SGB on Gold Returns :

$$R_{SGB} = 0.43 + 0.77R_{Gold}$$

- The coefficient of 0.77 implies that a 1% increase in gold returns leads to a 0.77% rise in SGB returns.
- The relationship is highly significant ($p < 0.001$) with $R^2 = 0.51$, suggesting that about 50% variation in SGBs price is explained by gold price movements.
- The remaining variation comes from interest income and lower liquidity.

Model 2: ETF on Gold Returns :

$$R_{ETF} = 0.07 + 0.88R_{Gold}$$

- The coefficient of 0.88 implies that a 1% increase in gold returns leads to a 0.88% rise in ETF returns.
- The relationship is highly significant ($p < 0.001$) with $R^2 = 0.74$, suggesting that about 74% variation in SGBs price is explained by gold price movements.
- This strong linear linkage highlights that ETFs act as an efficient market proxy for gold.
- Thus, ETFs are nearly perfect substitutes for physical gold in terms of price responsiveness, whereas SGBs are more stable.

4.3 ARCH Test

The ARCH test was conducted to check whether the returns of SGBs, Gold ETFs, and Physical gold exhibit volatility. The p-values for all three instruments (SGB = 0.33, ETF = 0.19, Gold = 0.87) are all above 0.05, indicating that we fail to reject the null hypothesis of homoskedasticity. This indicates no volatility clustering. Thus, extreme returns are isolated rather than persistent and the return fluctuations are relatively stable and random.

4.5 ADF Test (Stationarity)

All return series have p-values < 0.01, confirming stationarity at levels. This implies returns fluctuate around a constant mean and variance which is appropriate for regression and causality testing. Thus, the data are statistically stable and mean-reverting, suitable for time-series modelling.

4.6 Granger causality

The Granger causality results show that gold returns have a significant influence on SGB returns, particularly at a one-lag period ($p = 0.031$). This implies that changes in gold prices tend to predict short-term movements in SGBs, although with a slight delay. However, no such relation is found between gold and ETFs ($p > 0.75$ across all lags) which implies that ETFs react almost instantly to changes in gold prices due to their higher liquidity and active market trading.

4.7 Johansen Test

The Johansen test reveals that the trace statistics exceed critical values at all ranks, indicating at least one cointegrating relationship between the three instruments. This implies a long-run equilibrium relationship in which, despite short-term fluctuations, SGBs, ETFs, and gold prices move in the same direction over time. This means that even if SGBs or ETFs move differently for a short time, they eventually come back in line with gold prices. This shows that gold investments like SGBs and ETFs are now closely linked with global gold markets, helping India shift from using gold mainly for consumption to using it more as an investment.

5. Conclusion

The analysis reveals that SGBs, Gold ETFs, and physical gold all play unique but interconnected roles in India's gold investment scenario. Both SGBs and Gold ETFs derive their value from physical gold, the underlying asset, but they differ in how quickly and sharply they react to changes in gold prices. ETFs almost instantly mirror gold price movements, showing market efficiency and liquidity. SGBs, on the other hand, react with lag and show smoother price patterns, due to their fixed interest and lower trading frequency.

These findings show that SGBs bridge the gap between traditional and financialized gold by combining the security of physical gold with the potential for higher returns from financial instruments. Active investors use ETFs to track the market, meanwhile long-term savers use SGBs as a stable, income-yielding store of value. Thus , SGBs and ETFs have successfully transformed traditional gold savings into formal, market-driven, and efficient instruments for investment, which has decreased the dependence on imports thereby reducing the current account deficit. Thus, our study shows that SGBs combine the sheen of physical gold with the convenience and cost effectiveness of ETFs. They can, therefore, prove to be a useful tool for risk-averse long term investors and tax-efficient investment.

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